# Lecture 1: Linear Transformations & Matrices

Date: 2026-08-10
Reading: Chapter 2
Duration: 75 minutes
Difficulty: Intermediate

## Core Definitions

We begin by establishing the properties of fields mapping vector components.

### Definition 1.1 : Linear Map

A function $T : V \to W$ between two vector spaces over the same field $F$ is called a **linear transformation** if it satisfies:

1. Additivity: $T(u + v) = T(u) + T(v)$ for all $u, v \in V$
2. Homogeneity: $T(cv) = cT(v)$ for all $c \in F$ and $v \in V$

### Definition 1.2 : Vector Space

…

## Fundamental Theorems

### Theorem 1.3 : Rank-Nullity Theorem

Let $V$ and $W$ be vector spaces, where $V$ is finite-dimensional. If $T : V \to W$ is a linear map, then:

$$\dim(\text{null } T) + \dim(\text{range } T) = \dim V$$

## Homework Practice

### Exercise 1.4 : Identity Dimension Mapping

Let $T : \mathbb{R}^3 \to \mathbb{R}^2$ be defined by $T(x, y, z) = (x + y, z)$. Find a structured matrix representation $M_T$ and verify the output dimension explicitly:

$$M_T = \begin{pmatrix} 1 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}$$

### Exercise 1.5 : Identity Dimension Mapping

Let $T : \mathbb{R}^3 \to \mathbb{R}^2$ be defined by $T(x, y, z) = (x + y, z)$. Find the dimension of the null space.

**Solution:**

By inspecting the matrix row parameters:

$$M_T = \begin{pmatrix} 1 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}$$

The rank is clearly 2 because the two rows are linearly independent. By Rank-Nullity, $\dim(\text{null } T) = 3 - 2 = 1$.

### Note 1.6

Remember that every basis is linearly independent.

### Warning 1.7

Do **not** confuse image and codomain.

### Example 1.8

Let

$$T(x, y) = (x + y, y).$$

**Proof.**

We proceed by induction…

$\square$

### Remark 1.9

The converse is false.

### History 1.10

The Rank-Nullity theorem appeared in the nineteenth century.

### Lemma 1.11

lemma statement

**Proposition 1.12**

proposition statement

**Claim 1.13**

claim statement

**Corollary 1.14**

corollary statement